

Flynn's Classification of Computers

Q. Elaborate on Flynn's Classification of computer architectures. Describe SISD, SIMD, MISD, and MIMD with suitable examples.

[DEF] Definition to Remember

Flynn's Classification (1966) categorizes computer architectures based on the number of concurrent Instruction Streams (I) and Data Streams (D). The combinations yield four distinct models: SISD, SIMD, MISD, and MIMD.

1. The Four Categories of Flynn's Taxonomy

Acronym	Full Form	Concept & Working	Example
SISD	**Single Instruction, Single Data**	The traditional Von Neumann architecture	Early single-core PCs (e.g., Intel 808
SIMD	**Single Instruction, Multiple Data**	A single control unit issues the *same	**GPUs** (Graphics cards), Vector Proc
MISD	**Multiple Instruction, Single Data**	Multiple processors execute *different	Fault-tolerant systems (e.g., Space Sh
MIMD	**Multiple Instruction, Multiple Data**	Multiple processors fetch their own *i	**Modern Multi-core CPUs** (Core i7, R

2. Deeper Look at SIMD vs MIMD

* SIMD is highly efficient when the exact same operation needs to be applied to a massive array (e.g., adding brightness to a million pixels on a screen). All ALUs do the same math at the same time.

* M
Core 3

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Flynn's Taxonomy classifies architectures based on Instruction Streams (I) and Data Streams (D).
2. SISD: Single Instruction, Single Data. Traditional sequential architecture (Older PCs, 8086).
3. SIMD: Single Instruction, Multiple Data. Same operation on multiple data pieces (GPUs, Array Processors).
4. SIMD is the basis for Data-Level Parallelism.
5. MISD: Multiple Instruction, Single Data. Different operations on same data. Very rare (Fault-tolerant systems).
6. MIMD: Multiple Instruction, Multiple Data. Independent processors running independent tasks.
7. MIMD is the basis for Task-Level Parallelism (Modern multi-core CPUs like Intel i7).

[RECALL] Quick Recall

Flynn's Tax -> SISD (1 Instr, 1 Data, Old PCs) -> SIMD (1 Instr, Many Data, GPUs) -> MISD (Many Instr, 1 Data, Space Shuttles) -> MIMD (Many Instr, Many Data, Multi-core CPUs)